COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE 212 - Automating Server Management | **SECTION:** | CPE31S2 |
| **DATE PERFORMED:** | 09/10/26 | **DATE SUBMITTED:** | 09/10/26 |
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**ACTIVITY NO. 7.1** **Managing Files and Creating Roles in Ansible**

## 1. PROCEDURE OUTPUT

**Task 1: Create a file and copy it to remote servers**

1. Using the previous directory was created in your previous laboratory, create a directory, and name it “files.” Create a file inside that directory and name it “default_site.html.” Edit the file and put basic HTML syntax. Any content will do, as long as it will display text later. Save the file and exit.

```
erickalvarez@alvarezworkstation:~/CPE-212$ ls
ansible.cfg   install_apache_redhat.yml  README.md
combined.yml  install_apache.yml         roles
files         inventory.ini              site.yml
```

```
erickalvarez@alvarezworkstation:~/CPE-212$ ls files
default_site.html
erickalvarez@alvarezworkstation:~/CPE-212$
```

```
erickalvarez@alvarezworkstation:~/files$ cat default_site.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Web Server</title>
</head>
<body>

    <h1>Welcome to My Web Server</h1>

    <p>This webpage was deployed using ansible.</p>
    <p>This file was copied from the local repository to the remote server.</p>

</body>
</html>
```

2. Edit the site.yml file and just below the web_servers play, create a new file to copy the default html file for site:

```
  - name: copy default html file for site
    tags: apache, apache2, httpd
    copy:
      src: fileees/default_site.html
      dest: /var/www/html/index.html
      owner: root
      group: root
      mode: '0644'
```

3. Run the playbook site.yml. Describe the changes

```
PLAY RECAP *********************************************************************
192.168.56.105             : ok=5    changed=2    unreachable=0    failed=0
kipped=3    rescued=0    ignored=0
192.168.56.107             : ok=0    changed=0    unreachable=1    failed=0
kipped=0    rescued=0    ignored=0
192.168.56.108             : ok=8    changed=1    unreachable=0    failed=0
kipped=2    rescued=0    ignored=0

erickalvarez@alvarezworkstation:~/CPE-212$
```

```
erickalvarez@alvarezworkstation:~/CPE-212$ ssh ealvarez@192.168.56.108
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Sep 10 08:36:11 2026 from 192.168.56.111
[ealvarez@localhost ~]$ ls /var/www/html
index.html
[ealvarez@localhost ~]$ cat /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Web Server</title>
</head>
<body>

    <h1>Welcome to My Web Server</h1>

    <p>This webpage was deployed using ansible.</p>
    <p>This file was copied from the local repository to the remote server.</p>

</body>
</html>

[ealvarez@localhost ~]$ exit
logout
Connection to 192.168.56.108 closed.
erickalvarez@alvarezworkstation:~/CPE-212$
```

- After connecting to the managed nodes, we can see that after running the playbook, the HTML file was successfully copied to the managed nodes.

4. Go to the remote servers (web_servers) listed in your inventory. Use cat command to check if the

index.html is the same as the local repository file (default_site.html). Do both for Ubuntu and CentOS servers. On the CentOS server, go to the browser and type its IP address. Describe the output

Web_server:

```
erickalvarez@alvarezworkstation:~$ cat /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Web Server</title>
</head>
<body>

    <h1>Welcome to My Web Server</h1>

    <p>This webpage was deployed using ansible.</p>
    <p>This file was copied from the local repository to the remote server.

</body>
</html>

erickalvarez@alvarezworkstation:~$ s
```




**Welcome to My Web Server**

This webpage was deployed using ansible.

This file was copied from the local repository to the remote server.



- The CentOS default page was replaced by my HTML. This works since Ansible copied my HTML file to the /var/www/html/ directory as index.html, which Apache uses as the default webpage.

5. Sync your local repository with GitHub and describe the changes.

```
erickalvarez@alvarezworkstation:~/CPE-212$ git add .
erickalvarez@alvarezworkstation:~/CPE-212$ git commit -m "HOA 7.1"
[main 7ba0b3c] HOA 7.1
 Committer: erickalvarez <erickalvarez@alvarezworkstation.myguest.virtualbox.org
>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 2 files changed, 27 insertions(+)
 create mode 100644 files/default_site.html
```

```
erickalvarez@alvarezworkstation:~/CPE-212$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 808 bytes | 808.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects
To github.com:ErickAlvarezz/CPE-212.git
   54bcc3b..7ba0b3c  main -> main
```

- The changes include the new directory I added and the modified files that I uploaded to GitHub

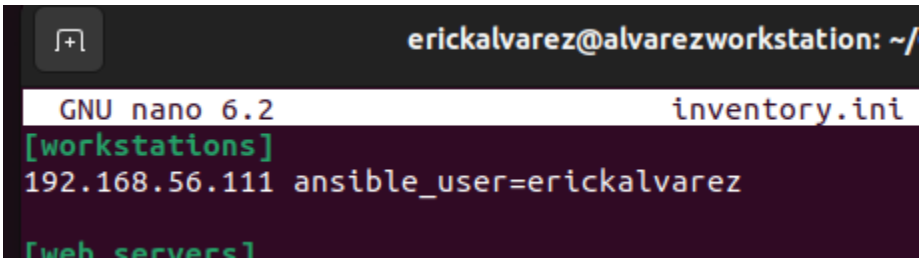
**Task 2: Download a file and extract it to a remote server**

1. Edit the site.yml. Just before the web_servers play, create a new play:

```
- hosts: workstations
  become: true
  tasks:
    - name: install unzip
      package:
        name: unzip
    - name: install terraform
      unarchive:
        src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28>
        dest: /usr/local/bin
        remote_src: yes
        mode: 0755
        owner: root
        group: root
```

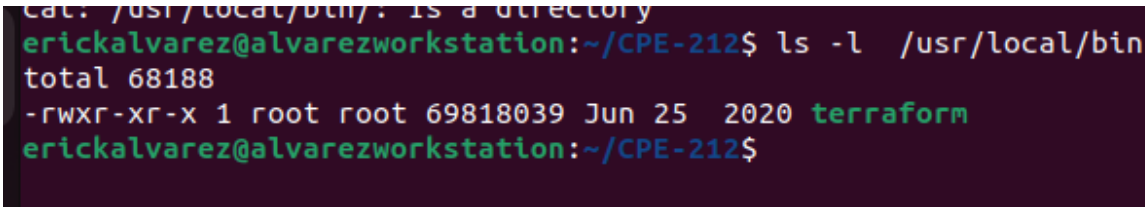
2. Edit the inventory file and add a workstation group. Add any Ubuntu remote server. Make sure to

remember the IP address



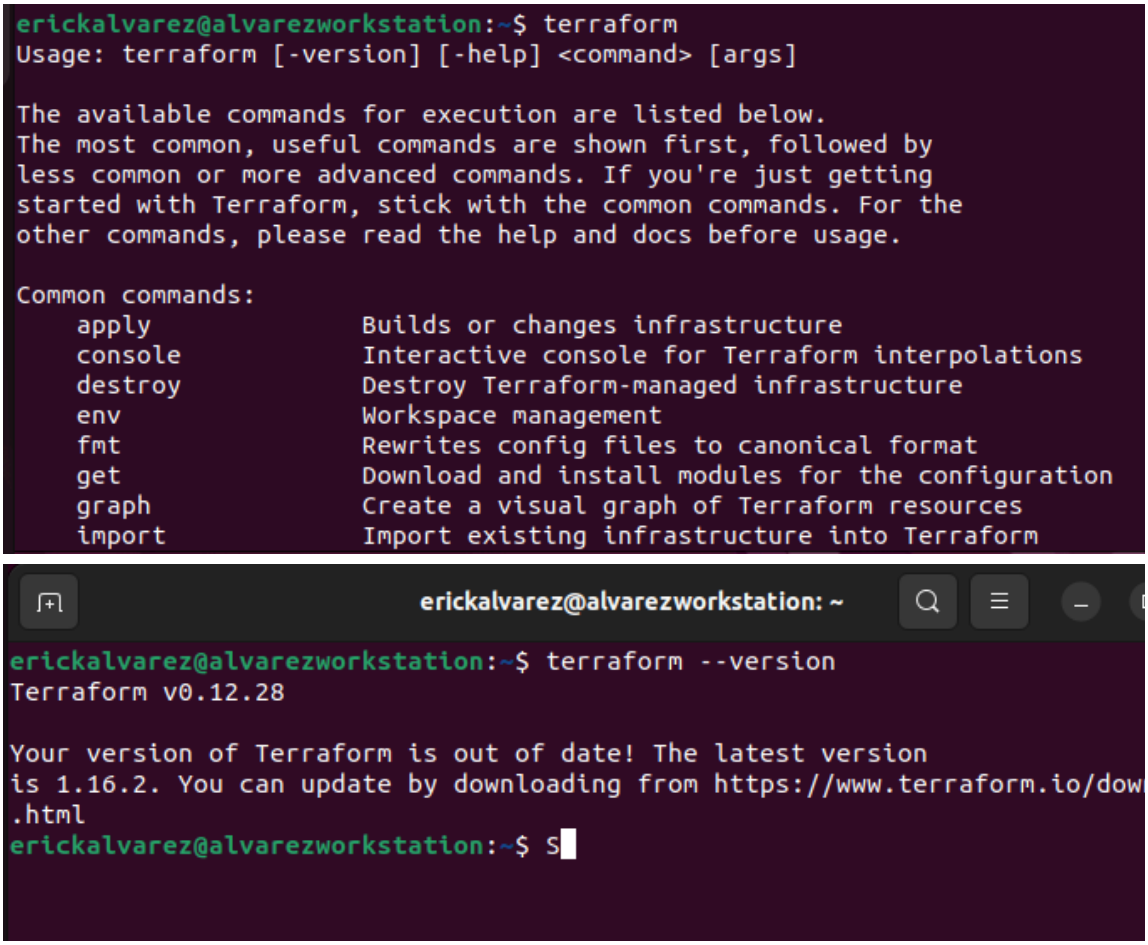

3. Run the playbook. Describe the output.




- We can see that the terraform has now been downloaded to the specified destination.

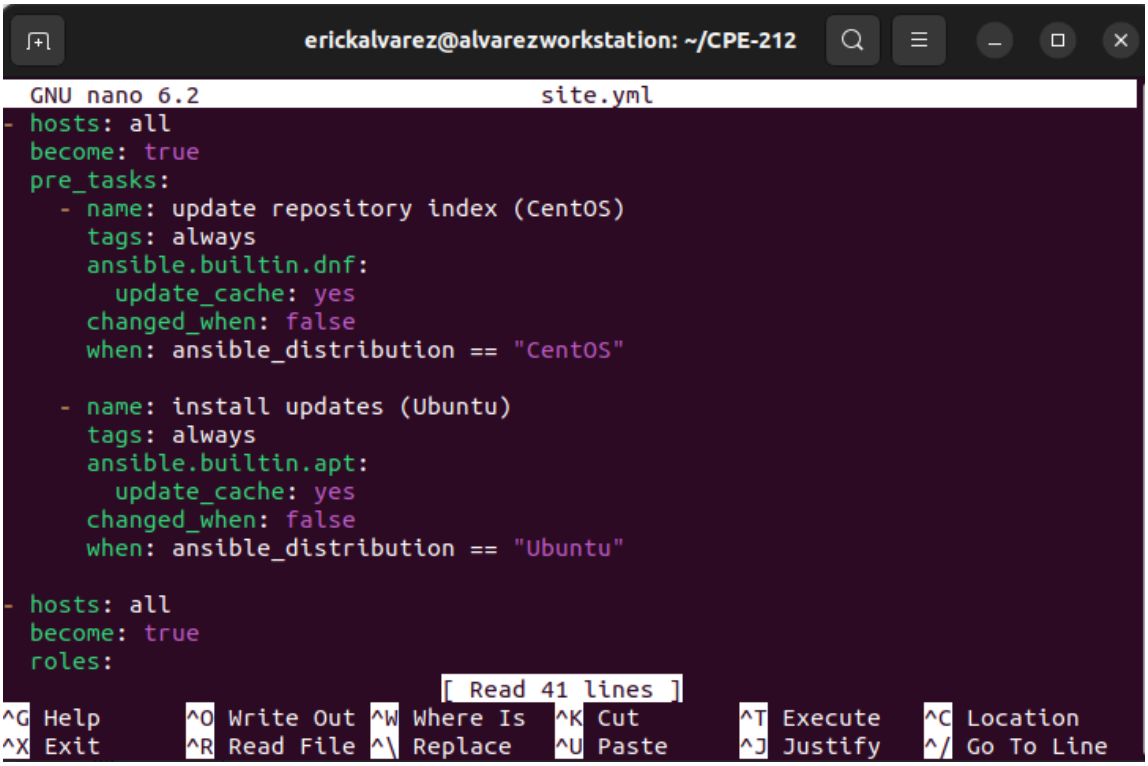
4. On the Ubuntu remote workstation, type terraform to verify installation of terraform. Describe the output.




- The output shows the built-in commands in Terraform. I also used terraform –version to verify that Terraform was successfully installed.

**Task 3: Create roles**

1. Edit the site.yml. Configure roles as follows: (make sure to create a copy of the old site.yml file because you will be copying the specific plays for all groups)




2. Under the same directory, create a new directory and name it roles. Enter the roles directory and create new directories: base, web_servers, file_servers, db_servers and workstations. For each directory, create a directory and name it tasks.

```
erickalvarez@alvarezworkstation:~/CPE-212/roles$ mkdir base
erickalvarez@alvarezworkstation:~/CPE-212/roles$ mkdir web_servers
erickalvarez@alvarezworkstation:~/CPE-212/roles$ mkdir file_servers
erickalvarez@alvarezworkstation:~/CPE-212/roles$ mkdir db_servers
erickalvarez@alvarezworkstation:~/CPE-212/roles$ mkdir workstations
erickalvarez@alvarezworkstation:~/CPE-212/roles$ ls
base  db  db_servers  file_servers  web  web_servers  workstations
erickalvarez@alvarezworkstation:~/CPE-212/roles$ s
```

```
erickalvarez@alvarezworkstation:~/CPE-212/roles/base$ ls
tasks
erickalvarez@alvarezworkstation:~/CPE-212/roles/base$
```

3. Go to tasks for all directory and create a file. Name it main.yml. In each of the tasks for all directories, copy and paste the code from the old site.yml file. Show all contents of main.yml files for all tasks

Base:

```
erickalvarez@alvarezworkstation:~/CPE-212/roles/base/tasks$ cat main.y
---
- name: install updates (CentOS)
  tags: always
  dnf:
    update_only: yes
    update_cache: yes
  when: ansible_distribution == "CentOS"

- name: install updates (Ubuntu)
  tags: always
  apt:
    update_cache: yes
  when: ansible_distribution == "Ubuntu"
erickalvarez@alvarezworkstation:~/CPE-212/roles/base/tasks$
```

Db_servers:

```
erickalvarez@alvarezworkstation:~/CPE-212/roles/base/tasks$ cd ~/CPE-212/ro
b_servers/tasks/
erickalvarez@alvarezworkstation:~/CPE-212/roles/db_servers/tasks$ cat main.y
---
- name: Install MariaDB on CentOS
  dnf:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: Start and enable MariaDB on CentOS
  service:
    name: mariadb
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"
```

File_servers:

```
erickalvarez@alvarezworkstation:~/CPE-212/roles/file_servers/tasks$ cat mai
---
- name: install samba package
  tags: samba
  package:
    name: samba
    state: latest
erickalvarez@alvarezworkstation:~/CPE-212/roles/file_servers/tasks$
```

Web_servers:

```
erickalvarez@alvarezworkstation:~/CPE-212/roles/web_servers/tasks$ cat main.yml
---
- name: Install Apache and PHP on Ubuntu
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: Install Apache and PHP on CentOS
  dnf:
    name:
      - httpd
      - php
    state: latest
```

Workstations:

```
erickalvarez@alvarezworkstation:~/CPE-212/roles/workstations/tasks$ cat mai
---
- name: install unzip
  package:
    name: unzip

- name: install terraform
  unarchive:
    src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28
x_amd64.zip
    dest: /usr/local/bin
    remote_src: yes
    mode: '0755'
    owner: root
    group: root
erickalvarez@alvarezworkstation:~/CPE-212/roles/workstations/tasks$
```

4. Run the site.yml playbook and describe the output.

```
PLAY RECAP *********************************************************************
192.168.56.105             : ok=7    changed=1    unreachable=0    failed=0
kipped=4    rescued=0    ignored=0
192.168.56.107             : ok=0    changed=0    unreachable=1    failed=0
kipped=0    rescued=0    ignored=0
192.168.56.108             : ok=10   changed=0    unreachable=0    failed=0
kipped=3    rescued=0    ignored=0
192.168.56.111             : ok=7    changed=1    unreachable=0    failed=0
kipped=2    rescued=0    ignored=0

erickalvarez@alvarezworkstation:~/CPE-212$
```

- After running the playbook, the recap shows that the tasks were completed successfully. Also it shows that I successfully created organized roles that works.

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of creating roles?
    - The importance of creating roles is to organize every role of each server. It's like a library with different genres in each section, you can easily find or modify) things that you want because everything is organized.
2. What is the importance of managing files?
    - Managing files makes tasks easier for the user. Tasks like copying, removing, moving, modifying, etc.

**Conclusions/Learnings:**

In conclusion, this activity helped me understand how to manage files and roles so that we do not have complex code for managing our servers. Thanks to this activity, this gave me an idea how to shortcut tasks when managing files, roles or even managing servers quickly in the real world.

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **N/A**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **N/A**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **N/A**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

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